Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/07/2018
PI Name:	Billman, Dorrit Ph.D.		
Project Title:	Training for Generalizable Skills & Knowledge: Integrating Principles and Procedures		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (3) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/06/2015	**End Date:**	07/05/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/5/2019 per NSSC information (Ed., 8/9/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Space Human Factors & Habitability (Ed., 1/19/17)		
Key Personnel Changes/Previous PI:	May 2017: no changes		
COI Name (Institution):	Catrambone, Richard Ph.D. (Georgia Tech Research Corporation)		
Grant/Contract No.:	NNX15AP26G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Understanding what properties of skills and knowledge (S&K) support useful generalization and long-term retention is critical to the success of future long-duration missions (e.g., DRM:Mars), yet there is a gap in our understanding of these properties (Space Human Factors Engineering-SHFE-TRAIN-04). Broadly, it is often the case that specialized procedural skills may be well-retained but difficult to generalize and abstract declarative knowledge may support generalization but is difficult to retain. Nevertheless, findings from the literature suggest that integrated knowledge is likely to be both generalizable and well retained. By integrated skills and knowledge we refer to ensembles of a) procedural skills and b) knowledge of principles that are integrated through relations such as explanation, instantiation, and associated use. We will conduct literature research followed by piloting and experimentation to a) define appropriate measures and manipulations and b) assess the extent that integrated S&K generalizes well and is well-retained relative to S&K covering related content but without the integrating relationships. Experimentation will aim to produce integrated knowledge though a variety of learning activities that a) build links from procedural skills to abstract principles through processes such as self-explanation, thus facilitating their decomposition and reuse and b) build links from abstract principles to specific procedures through processes such as instantiation, thus aiding their application. Evidence shows that integrated knowledge is better retained, as the many links of integrated knowledge provide multiple, alternative retrieval cues accessible in many contexts. We will assess whether manipulations that produce integrated knowledge also produce better generalization (as well as measuring retention), relative to knowledge that spans related content but is not integrated, and whether such effects can be attributed to the mediating role of integrated knowledge. We will use the work domain of operating and trouble-shooting complex technology, specifically spacecraft life-support systems as the test case for the proposed research. We will investigate “performing system-related tasks in highly autonomous environments” (Human Exploration Research Opportunities (HERO) Appendix A-25), such as operation, maintenance, and troubleshooting components within habitat life-support systems. Three factors motivate this choice.1) Such work is highly relevant to long-distance, crewed missions. 2) Prior research has studied operation of devices and thus provides findings on representations supporting generalization in similar contexts. 3) We will use an existing software suite including simulation of life-support devices on the International Space Station (ISS) and procedures for their operation as our test bed. This “micro-world” enables experimental control yet is a close analog of real mission work. In this domain the critical skills and knowledge are primarily cognitive (e.g., deciding what procedure to use), rather than sensory-motor (e.g., how hard a wrench must be turned). We include “meta-knowledge” such as identifying what information is missing when resources such as just-in-time-training are relevant. We will assess whether interventions related to integration produce forms of S&K that generalize well and are well retained, a prediction motivated by prior findings but not investigated directly, nor in NASA-relevant context. If integration is the basis for the predicted outcomes, it will provide a powerful principle for identifying and creating forms of S&K that are both generalizable and retainable. If not, we have discovered how learning is affected by important training interventions in a domain highly relevant to NASA future crewed exploration missions. This research will narrow the gap in understanding the factors that make skills and knowledge for NASA-critical tasks more generalizable and more retainable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Technical work across science, technology, engineering, and mathematics (STEM) disciplines, on Earth as in space, requires mastery of complex suites of knowledge and skills. Whether presented as education or as job-specific training, these skills and knowledge must be learned. Much of this work has an open, changing character such that even if time were available, it is not possible to anticipate and teach all the components that will be needed in the work. Thus, training that allows a person effectively to transfer skills and knowledge to situations and tasks that were not trained is extremely valuable. Indeed, transfer to different settings (from training environment to actual work) is often considered the primary success criterion for training; just a transfer from specific problem and procedures used in the classroom to novel situations is a critical goal of education. Understanding what types of knowledge and skills support transfer (and retention) will facilitate more effective training, for technical domains related to those we study.
	Year 3 Annual Report 2018- Training for Generalizable Skills & Knowledge: Integrating Principles and Procedures (#NNX15AP26G) Technical work across science, technology, engineering, and mathematics (STEM) disciplines, on Earth as in space, requires mastery of complex suites of knowledge and skills. Whether presented as education or as job-specific training, these skills and knowledge must be learned. Much of this work has an open, changing character such that even if time were available, it is not possible to anticipate and teach all the components that will be needed in the work. Thus, training that allows a person effectively to transfer skills and knowledge to situations and tasks that were not trained is extremely valuable. Our work investigates how to produce and to measure skills and knowledge that support significant generalization. Generalizing beyond the information and activities presented during training to novel goals, conditions, and resources is frequently a key training goal. However, the mental representations that best support this and the learning experiences most likely to produce these representations are not well understood. Our work studies how to train participants to enable them to generalize to novel situations and activities in technical work domains, specifically understanding and using equipment. In technical work domains of this sort generalization to new tasks not used in training is improved if the training provides experience both with the principles or model explaining how the technology works and with procedures for accomplishing domain goals with the technology. In working with equipment, it is valuable to learn both about how a device works and about the procedures or methods for using the device. Further, we claim that learning will be improved by training in which learners actively build relationships between a) the device model with its underlying principles and b) the procedures for operating the device. In particular, we propose that linking an understanding of the device with procedural skills for its operation will improve learners’ ability to generalize. Learners with integrated skills and knowledge of this sort will be better able to generalize their knowledge to new situations. They will be better able to accomplish work with goals, constraints, or resources different from than those encountered in training. For example, given training that only covered bringing up and powering down equipment, trainees with integrated training might be better able to generalize to a maintenance goal of testing a component. They

might also be better able to generalize to unexpected conditions in which equipment was initially configured differently than had been encountered in training. They might also be better able to predict or explain what will happening in a situation or how equipment works. From a methodological point of view, both design of training methods and of transfer measures are challenging. Many factors impact training and still more the application of what has been learned to the generalization tasks, including individual differences.

This year we continued research on training for generalization using the domain of habitat equipment on the International Space Station (ISS). We aimed to further develop methods and measures, in a conceptual replication and extension of Experiment 1, reported last year. Both Experiments 1 & 2 compared a training condition emphasizing integration (Integrating Condition) with a training condition that taught the same device model components and the same procedural skill components but did not emphasize linking the two types. Last year, we informally observed individual differences in attitudes about how the task should be approached. For example, some participants seemed more to expect that they would be directed quite specifically in what to do and that good performance would be close compliance, while others seemed more to expect they might need to be proactive, solve problems, or “think on their feet” and that good performance would include flexibility. We thought that the more active, flexible learning stance would facilitate linking of information and thinking about how this could be applied in unexpected situations. Therefore, we added an additional difference between conditions.

In the Experiment 2 Integrating Condition, we added training designed to instill into participants a take-responsibility, problem-solving attitude. Training sections were added that stressed the participant’s responsibility for safety of the International Space Station; the need to learn how systems, procedures, and NASA guidance are related; and the need to generalize knowledge to many situations. They were told that the best operators are active, alert, and integrate what they know. They were taught that NASA priorities place safety first, mission objectives next, then efficiency, and then consistency; and that the most desirable thing was to prevent problems, next to recover from problems, and then to report problems to Mission Control. As well as directly instructing users about the importance of these goals, users were confronted with problems they needed to address. These problems, however, were not in any of the ISS systems used in the training and transfer tasks. Instead, an unrelated Rover camera device and procedure was introduced and this completely different device was used to illustrate several types of example problems (e.g., the camera memory was full). Thus we aimed to train participants in the integrating condition to expect and to solve problems, without giving any exposure to problems they would encounter in the training domain.

For the comparison condition in Experiment 2, as in the Integrating Condition, participants were told the NASA priorities of safety, mission goal, efficiency, and consistency; and the problem management strategies of prevent, recover, and report. However this information was not repeated and integrated with other material and participants did not apply the problem management strategies. Thus, as with information about the device model and procedures, the same facts were presented, in the comparison as well as integrating condition, but they were not linked to practice nor to other information. However, rather than simply removing any higher level motivation or goal, we thought a more interesting comparison would be to provide this for the control condition as well, should we be able to identify and appropriate “neutral” goal or motivation. The best alternative motivation we identified was efficiency. Thus, the control condition emphasized the importance of fluent execution and getting as much done as possible given limited time; further, participants here had additional practice with strategies for efficiency, including use of the automaton software. This condition is labeled the Fluency Condition.

Task Progress:

We predicted that participants in the Integrating Condition would perform better on generalization tasks better than participants in the Fluency Condition; that this advantage would particularly be found for tasks less similar to those encountered in training; and that this would be found both for conceptual tasks and for operational tasks, particularly where operational tasks needed to draw on an understanding of how the device works.

Method. Our method is quite similar to that of Experiment 1, for which more detail is provided in the Task Book report for 2017. Concerning participants, we continued to use engineering students, primarily from astro-aero departments as our participants. Fifteen students participated in each condition.

Concerning domain, we continued to use the micro-world with simulated habitat equipment, including the Carbon Dioxide Removal System (CDRS), and the accompanying procedure execution software.

Concerning training, most of the materials and activity were quite similar to Experiment 1. We did expand ways that the Integrating and Fluency differed in integrating or separating procedural “how” information and model-based “why” information. Specifically, the Integrating Condition participants saw an animated functional model of the device, integrating how things worked with the device model, while the Fluency Condition participants saw an animated structural model of the device showing how components were made up of parts, to reduce the linkages to procedures suggested by the model. In addition, Experiment 2 provided different motivational emphasis, learning for flexibility and problem solving in the Integrating Condition versus learning for fluency.

Concerning measuring transfer, we modified several of our conceptual tasks to add more guidance on what we were looking for. We aimed to aid scoring and ensure participants were providing information on the topic intended. We added two operational tasks emphasizing fluency, to see if speed executing familiar tasks was all that was required and if participants in the Fluency Condition would perform better than in the Integrating Condition. Our coding was extended and revised on various tasks.

Results. Fortunately, most of our tasks are designed so that correctness and strategy are the primary measures rather than completion times. Due to a software problem, the time of computer actions, such as automatic execution of a procedure, increased very substantially over time, with the last third of users seeming to have considerably slowed performance. Therefore we have not prioritized tasks or measures dependent on software speed; we have not scored the two tasks designed to assess efficiency nor have we looked at completion times on any task. In addition, we have been investigating performance of the first cohort of 20 users, for whom the software operated closer to the expected speeds.

Scoring and analysis are still in progress, and findings are presented descriptively. Measures typically have large within-condition variability; condition differences are small on many measures; however, none of the measures scored to date suggest better performance in the Fluency Condition.

Several of the conceptual and operational tasks show patterns favoring the Integrating Condition. The quality of the narrative and diagram produced for the Conceptual Explanation task has been scored for the first 20 users. These Integrated Condition participants, mean of 26.6 9 (stdev = 3.4) points, scored notable higher than the Fluency

participants , mean of 18.6 (stdev=6.8), particularly on the narrative component. The Required Order task asked participants to list ordered action pairs from the CDRS Activation Procedure, where the earlier action must come before the later action, because of the way the CDRS works. Four pairs of "boxes" were provided for the participant to write in responses. Conditions did not differ in the number or correctness of pairs. They did, however, differ in diversity. Four pairs giving information about valves, fuses, water pump, and blower provides broader coverage than would a four pairs each stating a fuse must be closed before Valve 1...4 is opened. The Integrating Condition had a summed diversity score of 6.9 (stdev= .79) types of components and actions compared to the Fluency Condition score of 5.4 (stdev=1.6). This pattern is of note as the three diversity variables correlate significantly with condition, suggesting a real difference. Two other conceptual tasks showed no evidence of condition differences. One, the procedure writing task, had shown a performance advantage in Experiment 1; Integrating Condition participants had a median of 14 on a 16-point scale, compared to a Fluency Condition median of 5.5. In Experiment 2, both conditions had a median of 14, apparently near ceiling.

On the procedural tasks, three were designed to require reasoning about how the device works to solve correctly, Tasks D, E, and H. On these three the conditions required by the procedure are not met and the participant must reason about what other (unnamed) procedures must be run first to allow successful execution of the procedure that accomplishes the goal. If the participant reasons or understands the situation initially, the task can be accomplished without producing action failures, what we termed "preventive solutions." These are designed in increasing difficulty, where more indirect relations must be reasoned about. Conditions did not differ on the first of these problems, with performance pool in both (14% and 13% in Integrating and Fluency Conditions). Performance improved substantially on later problems (Task E & H) in the Integrating Condition, much more than in the Fluency Conditions (Task E: 46% vs. 13% and Task H: 43% vs. 21%). These differences are larger in the first cohort. A different type of operational task asked participants to open and close one fuse. This can be done by executing a small part of a startup and shutdown procedure. We were interested in whether participants formed the more precise method of executing only the relevant part of procedures or simply ran the whole procedure intact. In the Integrating Condition, 79% of users generated a selective strategy, versus 47% in the Fluency Condition.

Experiment 3.

We are conducting a retention study, in which participants in Experiments 1 or 2 return. We will assess their ability to carry out some of the same transfer tasks they did previously as well as some quite different tasks, about a Sabatier reactor. This will provide a window into retention of this material for durations of months to over a year.

Bibliography Type: Description: (Last Updated: 10/03/2019)

Papers from Meeting Proceedings

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